

CSSCM 2026 COHORT

EXERCISE 4

Dynamical Systems with NumPy: Pollutant Transport Through Connected Reservoirs

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Scenario

A water-treatment network contains five well-mixed reservoirs connected by pipes. A contaminant is accidentally introduced into Reservoir 1. As water moves through the network, contaminant is transported between reservoirs and removed from the system.

Let $C_1(t)$, $C_2(t)$, $C_3(t)$, $C_4(t)$, and $C_5(t)$ denote the contaminant concentrations in the five reservoirs at time t .

1. Coupled Dynamical System

The contaminant dynamics are described by:

$$\begin{aligned}dC_1/dt &= -0.40C_1 + 0.10C_2 \\dC_2/dt &= 0.20C_1 - 0.50C_2 + 0.10C_3 \\dC_3/dt &= 0.15C_2 - 0.45C_3 + 0.10C_4 \\dC_4/dt &= 0.10C_3 - 0.40C_4 + 0.10C_5 \\dC_5/dt &= 0.10C_4 - 0.25C_5\end{aligned}$$

Negative terms represent contaminant leaving or being removed from a reservoir. Positive terms represent contaminant entering from another reservoir.

2. Matrix Representation

Define the state vector:

$$C(t) = [C_1(t), C_2(t), C_3(t), C_4(t), C_5(t)]^T$$

The five coupled equations can then be written as:

$$dC/dt = A C$$

$$A = \begin{bmatrix} -0.40 & 0.10 & 0.00 & 0.00 & 0.00 \\ 0.20 & -0.50 & 0.10 & 0.00 & 0.00 \\ 0.00 & 0.15 & -0.45 & 0.10 & 0.00 \\ 0.00 & 0.00 & 0.10 & -0.40 & 0.10 \\ 0.00 & 0.00 & 0.00 & 0.10 & -0.25 \end{bmatrix}$$

3. Initial Condition

At $t = 0$, only Reservoir 1 is contaminated:

$$C(0) = [100, 0, 0, 0, 0]^T \text{ mg/L}$$

Use NumPy to create the matrix A and the initial state vector C .

4. Discretising the Dynamical System

Use the Forward Euler method. Approximate the derivative by:

$$dC/dt \text{ approximately equals } (C(n+1) - C(n)) / \Delta t$$

Substituting $dC/dt = AC$ and rearranging gives:

$$C(n+1) = C(n) + \Delta t [A C(n)]$$

Equivalently:

$$C(n+1) = (I + \Delta t A) C(n)$$

where I is the 5 x 5 identity matrix.

5. Numerical Simulation

Use a time step of **delta t = 0.1** and simulate from **t = 0** to **t = 20**.

At every time step calculate:

```
dCdt = A @ C
C_new = C + dt * dCdt
```

Store the concentrations from every time step in a two-dimensional array called `history`. Each row should represent a time and each column should represent a reservoir.

```
history = np.zeros((number_of_steps, 5))
```

Student Tasks - Core Simulation

1. Create the 5 x 5 matrix A and inspect its shape, `ndim` and `size`.
2. Create the initial concentration vector.
3. Generate the time values required for the simulation.
4. Create the 2D `history` array.
5. Store the initial concentration vector in the first row of `history`.
6. Use a loop and the Forward Euler equation to calculate the concentrations at later times.
7. Print the concentrations in all five reservoirs at $t = 0, 5, 10, 15$ and 20 .
8. Use slicing to extract the complete concentration history of Reservoir 1 and Reservoir 5.

6. Safety Threshold Challenge

The safety threshold for Reservoir 5 is:

$$C_{safe} = 5 \text{ mg/L}$$

9. Determine whether Reservoir 5 ever exceeds 5 mg/L during the simulation.
10. If it does, determine the first time at which the threshold is exceeded.
11. Determine the maximum concentration reached by each reservoir.
12. Determine the time at which each reservoir reaches its maximum concentration.
13. Identify which reservoir reaches its maximum concentration last.

Useful Array Hints

```
# Entire history of Reservoir 1
history[:, 0]

# Entire history of Reservoir 5
history[:, 4]

# Boolean condition
history[:, 4] > 5.0

# Maximum of each reservoir
np.max(history, axis=0)

# Index of maximum for each reservoir
np.argmax(history, axis=0)
```

7. Emergency Treatment: Piecewise Dynamical System

At $t = 8$, the treatment plant detects the contamination and activates an emergency purification unit in Reservoir 3.

Before treatment, the Reservoir 3 removal term is:

$$-0.45 C_3$$

After treatment begins, it becomes:

$$-0.90 C_3$$

The system therefore becomes piecewise:

$$dC/dt = A_1 C, \text{ for } t < 8$$

$$dC/dt = A_2 C, \text{ for } t \geq 8$$

Matrix A_1 is the original matrix. Matrix A_2 is identical except that the Reservoir 3 diagonal coefficient changes from -0.45 to -0.90.

Student Tasks - Emergency Treatment

14. Create A1 containing the original system matrix.

15. Create A2 by copying A1 and changing only the Reservoir 3 removal coefficient.

16. Repeat the simulation using an if/else condition to select the appropriate matrix.

```
if t < 8:
    dCdt = A1 @ C
else:
    dCdt = A2 @ C
```

17. Store this second simulation in a separate array called history_treatment.

18. Compare Reservoir 3 concentrations with and without emergency treatment.

19. Compare Reservoir 5 concentrations with and without emergency treatment.

20. Determine whether the treatment changes the maximum concentration reached in Reservoir 5.

21. Determine whether the treatment changes the time at which Reservoir 5 reaches its maximum.

8. Interpretation Questions

a. Why is this called a coupled dynamical system?

b. What physical meaning do the positive off-diagonal entries of A have?

c. What physical meaning do the negative diagonal entries have?

d. Why does changing only one element of A affect concentrations in reservoirs other than Reservoir 3?

e. Why must a numerical simulation use many successive time steps rather than solving only one matrix multiplication?

f. What effect might changing delta t from 0.1 to 1.0 have on the numerical approximation?

Expected Terminal Report

```
=====
EXERCISE 4 - RESERVOIR DYNAMICAL SYSTEM
=====
```

```
System matrix shape: _____
Number of simulation steps: _____
```

```
Concentrations at selected times:
```

```
t = 0 : [...]
t = 5 : [...]
t = 10 : [...]
t = 15 : [...]
t = 20 : [...]
```

```
Maximum concentrations:
```

Reservoir 1: _____
Reservoir 2: _____
Reservoir 3: _____
Reservoir 4: _____
Reservoir 5: _____

Reservoir 5 exceeds 5 mg/L: _____
First threshold crossing time: _____

Reservoir reaching its maximum last: _____

EMERGENCY TREATMENT

Maximum Reservoir 5 concentration
without treatment: _____

Maximum Reservoir 5 concentration
with treatment: _____

Change caused by treatment: _____
=====

Submission

Save your program as `exercise4_reservoir_system.py` and run it in Ubuntu using:

```
python3 exercise4_reservoir_system.py
```

Submit the Python script and a screenshot of the completed terminal output. All numerical values must be generated by the program. Do not submit manually calculated values in place of the simulation.